

Comparing Guide Agent Policies in Fire Evacuation Simulation: A Factorial Study of Reactive and Hybrid Deliberative Architectures

---

Ali Karaoglu

20587851

School of Computer Science
University of Nottingham

Abstract

Guide agent strategies for fire evacuation range from passive broadcasting to hybrid reactive-deliberative approaches. This project provides a systematic comparison of three such policies along an intelligence gradient: Beacon (passive broadcaster), Sweeper (reactive patrol), and Router (hybrid reactive-deliberative with congestion-aware routing). A factorial design of 14,400 simulation runs covers eight topologies, five fire positions, and three crowd sizes, analysed with one-way and two-way ANOVA and Tukey HSD pairwise comparisons. Every guide policy outperforms unguided evacuation by large margins, and the ranking is stable across every map tested: mean survival reaches 82.8% for Router, 78.4% for Sweeper, and 63.3% for Beacon, compared with 38.8% unguided. Router's advantage over Sweeper varies fourfold across topologies, growing largest on maps that force civilian flow through shared narrow passages while offering multiple exit routes. The policy ranking therefore generalises, while the practical size of Router's benefit is shaped by specific map geometry.

Declaration of AI Usage

For the report, Claude AI and Overleaf's built-in AI tools helped with proofreading, grammar and typo correction, tightening academic phrasing, providing feedback on section organisation, and translating my algorithm designs into LaTeX pseudocode formatting. During the research phase, Claude and ScienceDirect's built-in AI features were used to surface relevant literature and read article summaries.

For the codebase, AI tools helped with discussing algorithm choices, code review and debugging, suggesting inline comments and docstrings, efficiency advice (such as the multiprocessing worker count), and general implementation guidance.

For the experimental design and analysis, AI helped with the selection of appropriate statistical techniques, factorial-design discussions, and explanations of when and why each technique is appropriate.

All architectural decisions, agent and policy designs, code structure, experimental setup, statistical analysis, interpretation of results, and the content of equations, tables, and figures are my own work. AI outputs were reviewed by me and are not used as original work. I fully understand every part of this project and can explain and defend every decision made.

Contents

1 Introduction **4**
- 1.1 Motivation . . . 4
- 1.2 Research question . . . 4
- 1.3 Contribution . . . 4

2 Related Work **4**
- 2.1 Evacuation modelling approaches . . . 4
- 2.2 Guide and leader policies . . . 5
- 2.3 Congestion-aware routing . . . 5
- 2.4 Research gap . . . 5

3 Environment **6**
- 3.1 Building layouts . . . 6
- 3.2 Fire and smoke dynamics . . . 6

4 Agents **7**
- 4.1 Civilian agent . . . 7
- 4.2 Guide agent . . . 7
- 4.3 Beacon policy . . . 8
- 4.4 Sweeper policy . . . 8
- 4.5 Congestion-aware Router policy . . . 10

5 Implementation **12**
- 5.1 Technologies and architecture . . . 12
- 5.2 Challenges . . . 12

6 Experiments **13**
- 6.1 Experimental method . . . 13
- 6.2 Results . . . 13
- 6.3 Discussion . . . 17
- 6.4 Limitations . . . 18

7 Conclusion and Future Work **18**

References **19**

1 Introduction

1.1 Motivation

The effectiveness of a building evacuation depends both on the physical environment (layout, fire spread, exit capacity) and on how occupants are guided to exits. Trained leaders improve evacuation outcomes, and their strategies range from fixed-post broadcasting to actively guiding civilians around hazards. These strategies have rarely been compared directly under varied building conditions [1, 2], which motivates a systematic comparison of guide agent policies across a controlled range of topologies, crowd sizes, and fire positions.

1.2 Research question

This project addresses the following research question:

> *How do different intelligent guide agent policies affect fire evacuation outcomes, and under what building conditions does each policy perform best?*

This central question is divided into four sub-questions:

1. **Baseline effectiveness.** Does guided evacuation significantly outperform unguided?
2. **Policy ranking.** Which guide policy achieves the highest survival rate overall?
3. **Building topology.** Does building layout type (bottleneck versus open-plan) affect which policy is most effective?
4. **Crowd size.** How does crowd size affect the relative performance of guide policies?

1.3 Contribution

To address this question, three guide policies along an intelligence gradient are compared: Beacon (a passive broadcast agent), Sweeper (a reactive patrol agent), and Router (a congestion-aware hybrid reactive-deliberative agent). The three policies are evaluated across eight hand-designed building layouts and three crowd sizes, with five fire positions per map, giving a factorial design of 14,400 simulation runs. Statistical analysis uses one-way and two-way ANOVA with effect-size measures and Tukey HSD post-hoc comparisons to identify both the overall ranking and the conditions under which each policy performs best.

2 Related Work

2.1 Evacuation modelling approaches

Evacuation modelling is one of four research categories in the taxonomy of intelligent evacuation systems [1]. Research within this category has developed along three main lines; two covering pedestrian behaviour and a third covering how hazards spread.

Force-based models, introduced by Helbing and Molnár [3], treat each pedestrian as a particle subject to social and physical forces, producing emergent crowd behaviours such as lane formation and congestion at bottlenecks. Rule-based agent models take a different approach and give each pedestrian its own decision rules [4]. For example, Islas-Toski et al. [5] model pedestrians through five rules covering movement, stagnation, and three probabilistic behavioural states (accidents, hysteria, disorientation).

Cellular automata, by contrast, are commonly used for fire and smoke diffusion [6], dividing the environment into cells whose hazard density evolves through transition rules over neighbouring cells.

2.2 Guide and leader policies

Two complementary questions have driven agent-based research on trained evacuation leaders. One concerns how many leaders to deploy in a building. The other concerns what role they should play during evacuation.

McCormack and Chen [7] addressed the first by simulating building evacuation with varying proportions of trained leaders. They found that 5% of agents acting as leaders reduced evacuation time by around 40%, giving an optimal evacuee-to-leader ratio of 20:1. They also distinguished between leaders who guided from fixed posts and leaders who walked with the group, reporting a 25% time difference in favour of guiding from fixed posts. Pelechano and Badler [8] addressed the second by simulating trained leaders who communicated building route knowledge to other agents, finding that even small proportions of leaders substantially improved evacuation outcomes, particularly in layouts where ordinary agents could not identify exits independently.

2.3 Congestion-aware routing

The theoretical basis for congestion-aware routing lies in the speed-density relationship formalised by Hoogendoorn et al. [9], which demonstrates how crowd density reduces pedestrian speed and leaves civilians stuck at bottlenecks. Subsequent work has extended this insight into explicit routing algorithms.

Shikhalev et al. [10] proposed a multi-criteria cost function balancing path length, fire risk, and congestion. A follow-up study [11] applied this algorithm to shopping-mall layouts and reported 21-63% evacuation-time reductions over shortest-path routing. The same study found that the advantage vanished when no bottleneck was present. Wong et al. [12] reported similar gains of 24-35% through iterative route optimisation that updates paths based on ongoing crowd simulation.

Taking a different approach, Zhang et al. [13] formulated a density-weighted cost function that combines distance and crowd density in a single weighted sum,

$$c = \beta \cdot \mathrm{dist}(s.pos, p.pos) + (1-\beta) \cdot G_{\mathrm{density}}[s.pos], \tag{1}$$

where $\mathrm{dist}(s.pos, p.pos)$ is the Euclidean distance between the positions of nodes s and p, $G_{\mathrm{density}}[s.pos]$ looks up the density map at $s.pos$, and $\beta \in [0,1]$ balances distance against congestion. This formulation produces routes that prefer less congested corridors over shorter but more crowded ones.

2.4 Research gap

Two gaps in the literature converge on this work's research question. Evacuation path guidelines have received limited empirical comparison between guidance strategies [1]. Beyki et al. [2] applied statechart architecture [14] to a modular wildfire evacuation framework and independently identified dynamic rerouting under shifting hazard conditions as an underdeveloped area. This work addresses both gaps through a systematic comparison of three guide policies across varied building conditions.

3 Environment

3.1 Building layouts

Building topology strongly shapes evacuation dynamics: narrow corridors create queues at doorways, while open spaces allow but do not guarantee efficient flow to exits. Eight floor plans were designed across two categories. The bottleneck maps vary corridor width and exit placement: `bottleneck_A` is a grid of tight rooms with single-cell doorways; `bottleneck_B` widens these into multi-room corridors with spread exits; `bottleneck_C` places all four exits on one wall; and `bottleneck_D` is a hospital layout with a two-cell central aisle ending at exits at either end. The open-plan maps vary in layout: `open_plan_A` is a warehouse with scattered pillars; `open_plan_B` uses cubicle partitions, producing lane-like flow; `open_plan_C` funnels exits to a single edge; and `open_plan_D` splits into two zones joined by a narrow bridge. All maps are 36×80 cells at 10 world units per cell (≈ 3 m), giving a 108 m $\times$ 240 m footprint with four exits each. Holding scale and exit count constant isolates topology effects in between-map comparisons.

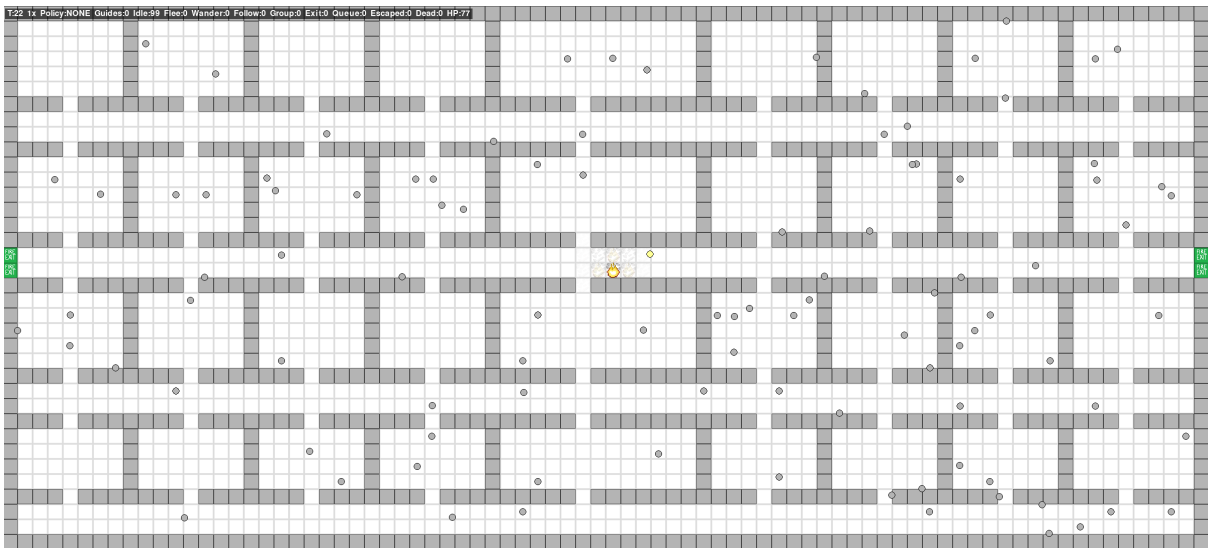


Figure 1. Visual mode running on `bottleneck_A`, unguided baseline.

3.2 Fire and smoke dynamics

Fire and smoke spread through the environment as coupled probabilistic cellular automata using Moore (8-connected) neighbourhoods [6]. At each tick, a cell's density ρ grows towards the maximum density among its neighbours, scaled by per-step rates $\alpha_{\text{fire}} = 0.005$ and $\alpha_{\text{smoke}} = 0.02$. Burning cells act as continuous smoke sources following the advection-diffusion model [15], which enforces $\rho_{\text{smoke}} \geq \rho_{\text{fire}}$ at every cell. Fires start at one of five positions (the centre or one of the four corners), giving 40 map-by-fire combinations that serve as experimental conditions.

The relative ordering of the two spread rates, with smoke outpacing flame, is consistent with corridor smoke-front measurements that exceed early-stage flame-spread velocities [16, 17]. Their absolute magnitudes deliberately compress real-world fire-hazard timescales so that hazardous conditions develop within the 6000-tick ($\approx$200-second) simulation horizon.

4 Agents

4.1 Civilian agent

Each civilian is a reactive agent in the sense of Brooks [18]. It observes local features and follows behavioural rules, without internal planning. This keeps individual behaviour simple and delegates coordination to the guide policies rather than the civilians.

Civilians pass through eleven states that match the phases of a real evacuation. They begin unaware, freeze briefly after detecting danger (a 15–60-tick pre-evacuation delay following Proulx [19]), then move in one of six modes: room-fleeing, wandering, wall-following in heavy smoke, crowd-herding, following a guide's group, or moving to a known exit. The sequence ends in queueing at an exit, escape, or death.

Following the rule-based approach common in evacuation modelling [4, 5], five behavioural rules drive the agent: random walking without a goal, BFS pathfinding when one is known, smoke-induced disorientation, herding on visible crowd motion [20], and potency-based damage [21]. Each civilian has a potency value (a health-like resilience measure) initialised uniformly in $[50, 100]$, modelling individual variation in hazard endurance. At each tick, potency decreases by

$$\Delta p = 3\rho_f + \rho_s, \tag{2}$$

where ρ_f and ρ_s are the local fire and smoke densities. The two terms reflect burns and smoke inhalation as separate effects. Perception is similarly smoke-dependent, with separate ranges for people (70 world units [22]) and exit signs (25 world units [23]). Walls block the line of sight in both cases.

Crowd-level coordination draws on Reynolds' boids model [24], originally designed for free-space flocking of birds and fish but adapted here for a walled indoor environment. A universal separation force prevents overlap between civilians. Within a guide's group, a BFS-based cohesion rule lets civilians follow the guide around corners, while boids alignment keeps their velocity matched to the guide's.

4.2 Guide agent

All guides take fire and smoke damage when local density exceeds 0.3. At full density, fire becomes fatal after 150 ticks and smoke after 500, making fire the dominant threat. When a guide's health drops below 60%, it abandons its current task and sprints to the nearest exit. The 60% reserve leaves enough health for the exit route, since by the later stages of the simulation fire and smoke typically cover most routes to the exits.

Sweeper and Router, the two escort-capable policies, recruit into the guide's group any civilian within a smoke-attenuated visual range [22]. Group size is capped at a dynamic ceiling m that balances collection efficiency against exposure time. Implemented as zero-order Takagi–Sugeno fuzzy inference [25, 26], it simplifies to a linear interpolation between a base value and a floor,

$$m = \max(f,\ b - d \cdot (b - f)), \tag{3}$$

where $b = \text{round}\left(3\sqrt{n/g}\right)$ is the base value for n civilians and g guides, $f = 5$ is the floor, and $d \in [0, 1]$ is the normalised local danger given by $\max(\rho_f/0.1, \rho_s/0.4)$. During an escort the ceiling is lifted, so any civilian seen along the route is folded into the group. Escort paths use a three-layer cascade (Algorithm 1): hazard-penalised Dijkstra favours cleaner routes; a safety validator rejects any path with a waypoint above the 0.3 danger threshold; and BFS provides a last-resort walkable path when no Dijkstra candidate survives. Guides walk at $3/4$ of their solo speed while carrying a group, to keep pace with civilians. On arrival, civilians within a few cells receive a direct exit target, transitioning to the exit queue. A release timeout ensures any straggler is handed an exit target rather than abandoned.

Algorithm 1 Three-layer escort-path cascade (shared by Sweeper and Router)

function PLANESCORTPATH(*env*, *start*, *goal*, *fire*, *smoke*)
  $danger \leftarrow \max(fire, smoke)$ ▷ element-wise per cell
  $path \leftarrow$ DIJKSTRAWITHHAZARDPENALTIES$(env, start, goal, fire, smoke)$
  if $path \neq \varnothing$ **then**
    $safe \leftarrow$ **true**
    for all waypoint w in $path$ **do**
      if $danger[w] > 0.3$ **then**
        $safe \leftarrow$ **false**; **break**
      end if
    end for
    if $safe$ **then return** $path$
    end if
  end if
  return BFS$(env, start, goal, danger)$
end function

4.3 Beacon policy

As a passive guide policy, each Beacon guide broadcasts the location of its assigned exit to nearby civilians. Each broadcaster stays two cells inward from its exit to avoid blocking it, and reaches every civilian within audible range.

At each tick, civilians within range hear the broadcast with a probability that decreases with distance and with smoke at both the listener and the source. The 30-metre range follows megaphone acoustics in fire conditions (Table 1): a 95 dB source loses 6 dB per doubling of distance, reaching about 65 dB at 30 m, just above the 60–70 dB background. A linear distance falloff is used in place of the inverse-square law, a simplification that keeps the broadcast useful across its full range while retaining monotonic decay.

Table 1. Megaphone attenuation at selected distances.

Distance (m)	Level (dB)
1	95
2	89
4	83
8	77
16	71
30	65

An unaware civilian wakes with a shortened reaction delay and then heads for the announced exit. A civilian already evacuating is redirected there immediately. In both cases, Beacon provides information, not escort. Once told the exit location, civilians find their own way.

4.4 Sweeper policy

As a reactive guide policy [18], Sweeper represents rule-based response to local conditions. Each guide is assigned a territory, cycling through four phases (Algorithm 2): patrol to collect civilians, escort them to the nearest safe exit, release them on arrival, and return for more. Exit choice is strictly local, picking the nearest safe exit. Each candidate exit x receives a score

$$s_x = d_x + \beta \cdot \rho_{s,x} + \gamma \cdot \rho_{f,x}, \tag{4}$$

where d_x is the Euclidean distance from the guide to exit x, $\rho_{s,x}$ and $\rho_{f,x}$ are the current smoke and fire densities at the exit, with the smoke weight $\beta = 200$ and the fire weight $\gamma = 500$. The weighting reflects hazard severity at the destination. Fire is assigned the higher weight as the more immediate threat, followed by smoke, which remains harmful upon arrival.

Algorithm 2 Sweeper phase loop (one tick)

```
procedure SWEEPERSTEP(env, fire, smoke, civilians)
    BASE.UPDATE(env, fire, smoke, civilians)
    if not alive or self_evacuating then return
    end if
    UPDATEBLACKBOARD(position, patrol_progress)
    if phase = PATROL then
        MOVETONEXTPATROLPOINT()                          ▷ alternate above/below spawn
        RECRUITVISIBLE(civilians)
        if emergency or |group| ≥ EFFECTIVEMAXGROUP() then
            phase ← ESCORT
            CHOOSENEARESTSAFEEXIT()
            PLANESCORTPATH()                                          ▷ Algorithm 1
        else if territory cleared then
            REALLOCATEFROMBUSIESTPEER()                          ▷ workload channel
        end if
    else if phase = ESCORT then
        if ARRIVEDATEXIT() then phase ← RELEASE
        end if
    else if phase = RELEASE then
        HANDCIVILIANSTOEXITQUEUE()
        phase ← RETURN
    else if phase = RETURN then
        RETURNTOTERRITORY()
        if ARRIVED() then phase ← PATROL
        end if
    end if
end procedure
```

Patrol points are drawn from two sources: room centres where civilians typically start, and a grid covering the corridors between them. Rather than visiting them nearest-first, the guide alternates between points above and below its starting position. This interleaving ensures both halves of the territory are covered in parallel, rather than one side being cleared before the other is reached. At each point the guide pauses to let civilians come within visual range, recruits nearby civilians, and decides whether to continue collecting or begin escorting. The shared commitment ceiling samples current fire and smoke. A 2-cell scan for fire above 0.1 or smoke above 0.4 (or smoke above 0.3 on a group member) forces an immediate escort; these thresholds match the civilian fire-damage onset and smoke-disorientation threshold.

The four guides split the map into quadrants, each patrolling its own zone, so their patrol effort covers the building in parallel rather than converging on the first hotspot. Coordination between guides runs through a shared blackboard with four reactive channels, carrying each guide's current position, currently claimed patrol target, locations already cleared, and uncleared patrol count. When a guide has cleared all its patrol points, it reads the workload channel to identify the busiest peer and adopts half of that peer's uncleared points, mirroring fire-brigade radio coordination on the ground. The patrol-target and cleared-cell channels prevent two guides from walking to the same cell after reallocation. If no peer has enough work to share, the guide resumes patrolling its own territory in case wandering civilians have drifted into previously empty cells.

4.5 Congestion-aware Router policy

Router extends reactive architecture [18] with a short-horizon forecast of hazard spread, creating a hybrid reactive-deliberative approach. Each Router scans the map globally, cycling through five phases (Algorithm 3): scan for hotspots, respond to a chosen one, collect civilians, escort them to an exit, and release them on arrival before resuming the scan. Exit choice is deliberative, scoring each exit on distance, predicted congestion, and predicted danger, so the guide can prefer a farther but cleaner exit over a nearer but crowded one.

Algorithm 3 Router phase loop (one tick)

procedure ROUTERSTEP(*env*, *fire*, *smoke*, *civilians*)
  BASE.UPDATE(*env*, *fire*, *smoke*, *civilians*)
  if not *alive* or *self_evacuating* **then return**
  end if
  UPDATEBLACKBOARD(*position*, *tick*)
  predicted_fire, *predicted_smoke* ← PREDICTSPREAD(30)
  if *phase* = SCAN **then**
    hotspot ← FINDHOTSPOT(*civilians*)
    if *hotspot* ≠ ∅ **then**
      phase ← RESPOND; PLANPATH(*hotspot*)
    end if
  else if *phase* = RESPOND **then**
    if ISDANGEROUS(*hotspot*) **then**
      phase ← SCAN
    else if ARRIVED() **then**
      phase ← COLLECT
    end if
  else if *phase* = COLLECT **then**
    RECRUITVISIBLE(*civilians*)
    if emergency **or** |*group*| ≥ EFFECTIVEMAXGROUP() **then**
      phase ← ESCORT
      CHOOSEBESTEXIT()
      PLANESCORTPATH() ▷ Algorithm 1
    end if
  else if *phase* = ESCORT **then**
    if *tick* mod 20 = 0 **and** not in commitment zone **then**
      REEVALUATEEXIT()
    end if
    if ARRIVEDATEXIT() **then** *phase* ← RELEASE
    end if
  else if *phase* = RELEASE **then**
    HANDCIVILIANSTOEXITQUEUE()
    phase ← SCAN
  end if
end procedure

Hotspots are identified by DBSCAN [27] over civilian positions, finding clusters of arbitrary shape without requiring a preset number of clusters. The cluster centroid becomes the target, snapped to the nearest walkable cell if it lands on a wall. After three successive empty scans, the search falls back to the nearest isolated civilian so late-stage stragglers are not abandoned. If the target hotspot becomes dangerous between scanning and arrival, the guide abandons it and rescans, prioritising safer hotspots where more civilians can be reached. Path planning combines density-weighted Dijkstra [13] over a 3×3 blurred density grid with the shared safety cascade, so civilians raise the cost of neighbouring cells and shape the cost landscape around local crowds. Extending Zhang et al.'s formulation with additional fire and smoke penalties, the per-cell cost is

$$c_e = 1 + w_f \cdot \rho_f + w_s \cdot \rho_s + w_d \cdot \rho_d, \tag{5}$$

where ρ_f, ρ_s, and ρ_d are the fire, smoke, and civilian-density values at the destination cell. Reflecting hazard severity, fire receives the highest weight ($w_f = 20$) as the most lethal hazard, followed by civilian density ($w_d = 8$) for its congestion effect, and smoke ($w_s = 5$) which remains passable.

The deliberative element is a 30-tick forward simulation of the cellular automaton (roughly matching an escort leg), computed deterministically. Exit scoring, path costs, and the shared group-size ceiling all operate on this predicted hazard. During escort, predicted congestion and danger can shift, so the exit is re-evaluated every 20 ticks. Each candidate exit x receives a score

$$s_x = d_x + \alpha \cdot (q_x + i_x) + \beta \cdot \rho_{s,x} + \gamma \cdot \rho_{f,x}, \tag{6}$$

extending Sweeper's local scoring (Eq. 4) with an additional congestion term, and replacing the current hazard densities with values predicted 30 ticks ahead. The queue q_x is the current number of civilians queuing at exit x, i_x is the sum of group sizes of other Routers heading there, and $\alpha = 30$ weights the combined count. The congestion weight is modest relative to β and γ, so Router can accept a slightly busier exit when it is much cleaner. The guide switches to any candidate that scores at least 30% better than the current target.

Re-evaluation is suppressed within a five-cell commitment zone of the current exit. Without it, a switch triggered close to the target could turn the guide around before release. Since civilians escape only on release, not on crossing an exit cell, they would walk past the near exit and accumulate smoke damage on the detour. After any emergency reroute, the guide enters a ten-tick cooldown before further rerouting is allowed, preventing thrashing when the danger grid keeps changing. If waypoints run out mid-escort, away from any exit, a fresh path is recomputed through the same safety cascade rather than leaving the guide immobile.

Inter-guide coordination runs through a shared blackboard with three channels: hotspot-claims, positions, and exit-claims. Hotspot claims prevent two Routers from targeting the same cluster. Position updates feed a cumulative inverse-distance repulsion. Adapting Reynolds' separation force [24], this bias pushes newly chosen hotspots away from regions where peers are working, spreading the guides across the map. Exit claims carry the group size of each ongoing escort. Other Routers count incoming arrivals in their own exit scoring, which prevents multiple guides converging on the same exit. Router therefore bases its decisions on current observations, forward prediction, and peer coordination, placing the policy beyond Brooks' reactive baseline.

5 Implementation

5.1 Technologies and architecture

The simulation is implemented in Python 3 using the libraries summarised in Table 2.

Table 2. Python libraries used in the simulation and the analysis pipeline.

Library	Purpose
NumPy	Grid operations and random number generation
Pygame	Real-time visualisation (visual mode only)
Pillow	Animated GIF frame loading (visual mode only)
matplotlib	Analysis figures
pandas	Tabular data handling for analysis
scipy	One-way ANOVA and effect-size computation
statsmodels	Tukey HSD post-hoc and two-way ANOVA
zlib	Cross-process seed determinism

The codebase is organised as a package with per-policy modules, as shown in Figure 2.

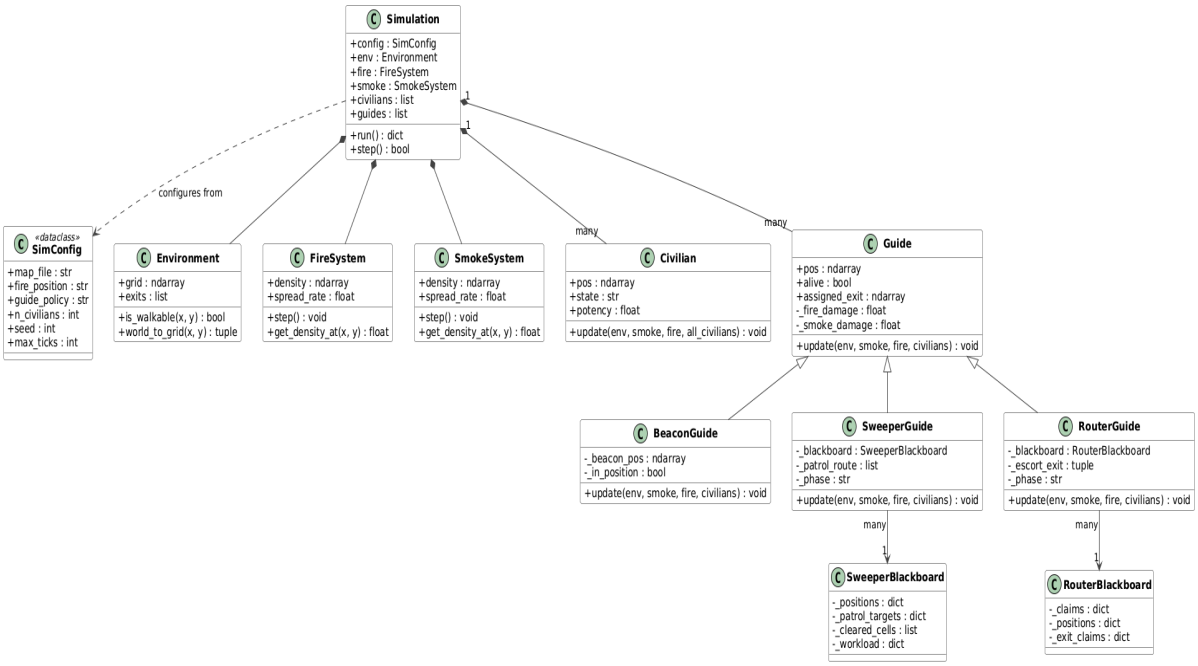


Figure 2. Class hierarchy of the simulation

5.2 Challenges

Router's deliberative planner forecasts fire and smoke 30 ticks ahead, but the cellular automaton is stochastic. A faithful Monte Carlo forecast (4 guides $\times$ N samples $\times$ 30 steps $\times$ 36×80 cells per simulation tick) is computationally prohibitive. The forecast therefore replaces the per-cell noise term with its expected value, yielding a deterministic single-pass approximation of the stochastic mean.

The original seed generator applied Python's built-in `hash()`. Because `hash()` is randomised per process by default, a fresh process produced different seeds for the same condition, breaking cross-session reproducibility. Replacing `hash()` with `zlib.crc32`, which is deterministic across processes and Python versions, fixed this.

6 Experiments

6.1 Experimental method

The experiment uses a fully factorial design with eight maps, five fire positions, four guide policies, three crowd sizes, and thirty trials per condition (deterministic seeds), giving 14,400 runs in total. The four guide policies are Beacon, Sweeper, Router, and an unguided baseline in which no guide is spawned. Each guided condition spawns four guides per map, so the three tested crowd sizes of 50, 100, and 150 civilians correspond to guide-to-civilian ratios of 1:12.5, 1:25, and 1:37.5, spanning well-supplied, near-optimal, and stressed conditions.

The primary outcome is survival rate, defined as the proportion of civilians who escaped. Four additional metrics follow Kasereka et al. [21], comprising TV (count escaped), TM (count deceased), MP (mean potency of survivors), and MT (mean escape time). Escape counts are also binned into twelve 500-tick windows across the 6,000-tick simulation horizon, supporting cumulative evacuation curves. This horizon was set from worst-case measurement: smoke-coverage time across the forty map-by-fire scenarios, plus a buffer for civilian death time and safety margin.

Statistical analysis uses one-way ANOVA on the policy main effect with η^2 as the effect-size measure. Two-way ANOVAs test the interactions between policy and map type (bottleneck versus open-plan), and between policy and crowd size. Pairwise comparisons use Tukey HSD post-hoc tests with Cohen's d. Tukey HSD applies family-wise correction across the six pairwise policy comparisons; the eight per-map paired tests (Table 5) are reported uncorrected, but all p-values fall below the Bonferroni-corrected family-wise threshold of $\alpha/8 = 0.00625$. At approximately 3,600 observations per policy group, the Central Limit Theorem supports ANOVA robustness against mild violations of normality and homoscedasticity.

6.2 Results

The full factorial of 14,400 runs completed without incident: no run reached the 6,000-tick cap, no civilian was unresolved at end, and no policy errored. Survival rate therefore captures the final fate of every civilian without missing data.

Policy emerged as the dominant factor on this outcome. Mean survival increased monotonically across the four policies: from 38.8% unguided through 63.3% for Beacon, 78.4% for Sweeper, and 82.8% for Router (Figure 3). One-way ANOVA confirmed the separation at $F(3, 14396) = 2457.59$, $p < 0.001$, with $\eta^2 = 0.34$ indicating a large effect size for policy alone. Pairwise Tukey HSD contrasts (Table 3) were significant for every policy pair at $p < 0.001$, with effect sizes ranging from a small $d = 0.26$ between Router and Sweeper to a very large $d = 1.79$ between Router and the unguided baseline.

Table 3. Tukey HSD pairwise comparisons on survival rate. Mean differences are oriented so the higher-ranked policy appears second. All pairs significant at $p < 0.001$.

Pair	Mean diff.	Cohen's d	Magnitude
none vs Beacon	+0.246	0.83	large
none vs Sweeper	+0.397	1.59	large
none vs Router	+0.441	1.79	very large
Beacon vs Sweeper	+0.151	0.64	medium
Beacon vs Router	+0.195	0.84	large
Sweeper vs Router	+0.044	0.26	small

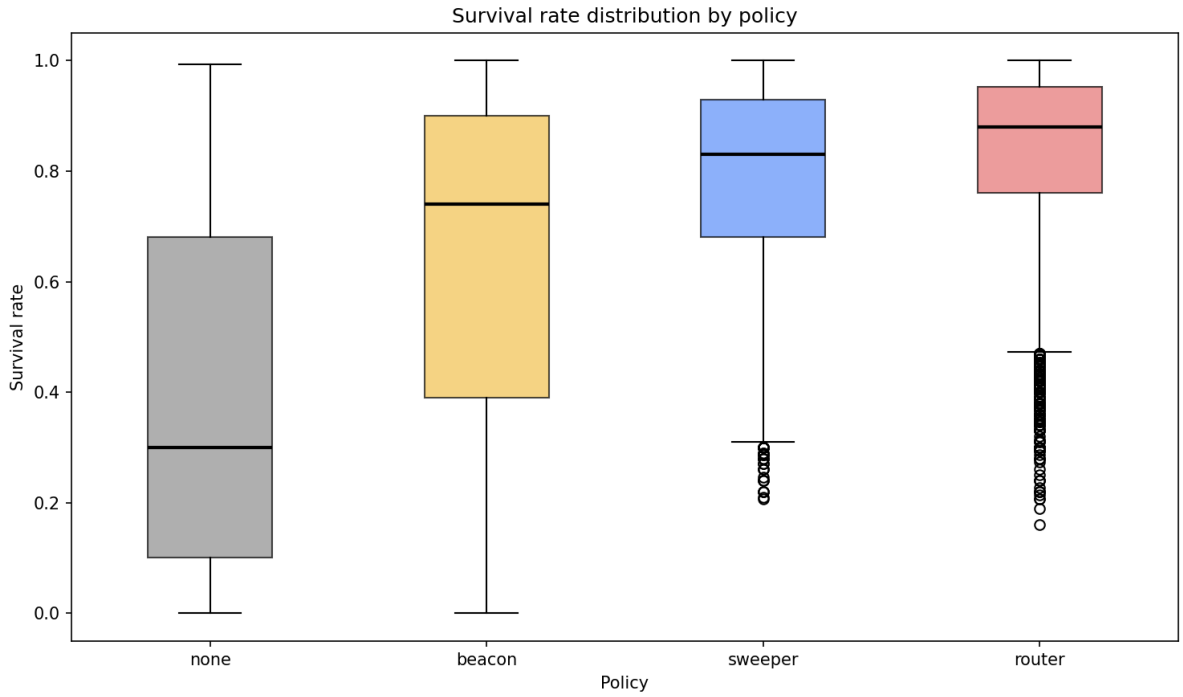


Figure 3. Survival rate distributions by guide policy across all 14,400 runs. Boxes show the interquartile range with the median as a horizontal line.

Although the policy ranking is identical on every map, the gap between policies varies considerably across conditions. Three two-way interactions isolate this (Table 4). Policy × map-type shows different policy gaps between bottleneck and open-plan maps (Figure 4); policy × crowd-size shows near-flat curves for Router and Sweeper against rising curves for Beacon and the unguided baseline (Figure 5); and policy × fire-position is negligible. The ranking therefore holds everywhere, even where the gaps vary.

Table 4. Two-way ANOVA results for survival rate, with partial η^2 effect sizes. All terms significant at $p < 0.001$ except policy × fire position ($p = 0.024$).

Term	F	Partial η^2	Magnitude
Model: policy × map type			
Policy (main)	3930	0.450	large
Map type (main)	7863	0.353	large
Policy × Map	256	0.051	small
Model: policy × crowd size			
Policy (main)	2496	0.342	large
Crowd size (main)	41	0.006	negligible
Policy × Crowd	25	0.010	small
Model: policy × fire position			
Policy (main)	2501	0.343	large
Fire position (main)	62	0.017	small
Policy × Fire	2.0	0.002	negligible

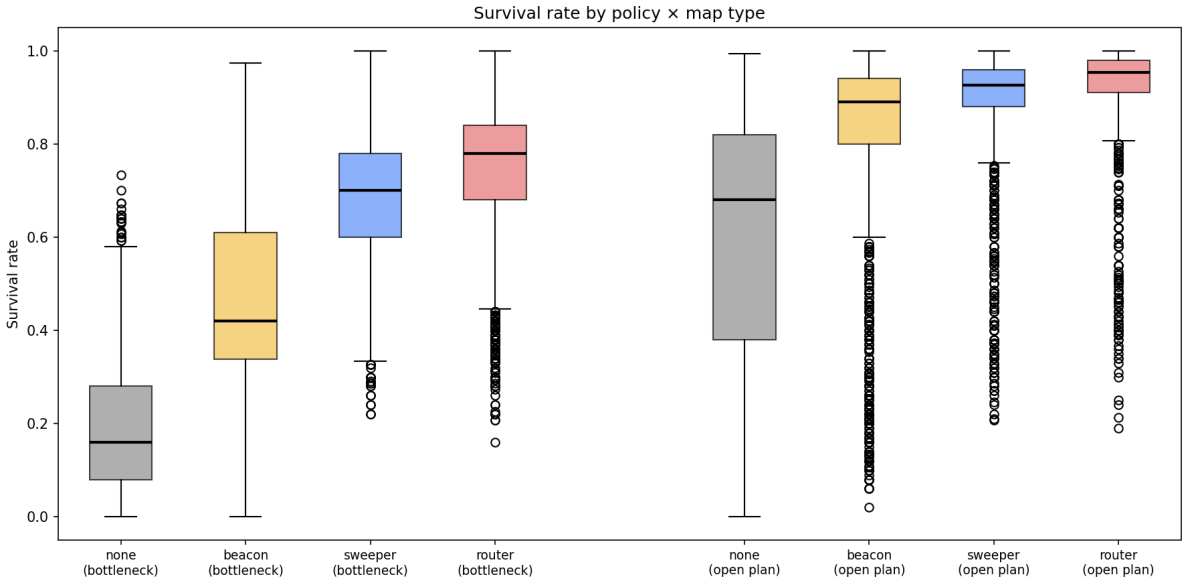


Figure 4. Survival rate distributions split by map type (bottleneck or open-plan) and policy.

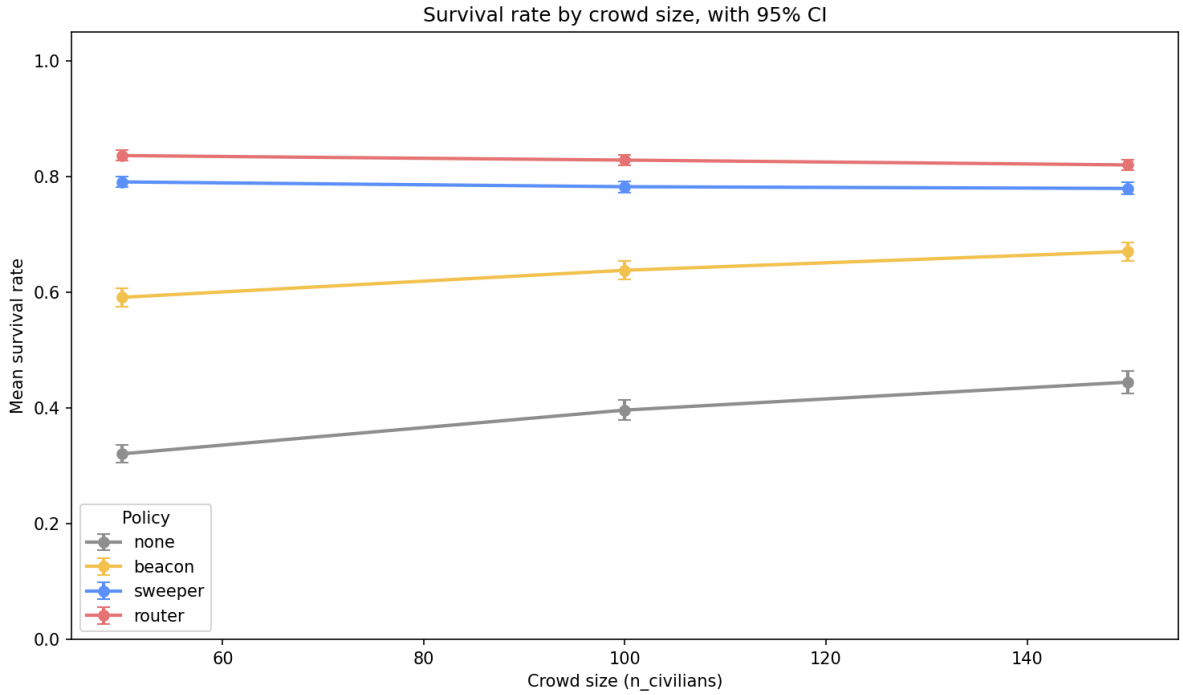


Figure 5. Mean survival rate as a function of crowd size, one line per policy. Error bars are 95% confidence intervals.

The topology interaction is most visible in the per-map breakdown of Router against Sweeper (Table 5). Router achieved the highest survival rate on all eight maps (Figure 6), but the size of its lead varied fourfold, from $d_z = 0.29$ on `bottleneck_C` to $d_z = 1.17$ on `bottleneck_D`. Six maps fell in the small-effect range ($d_z < 0.5$), and two (`bottleneck_A` and `bottleneck_D`) in the large-effect range ($d_z > 0.8$), with no maps in the medium range. Excluding these two maps from the two-way ANOVA reduces the policy × map-type interaction's partial η^2 from 0.051 to 0.015 (a 70% reduction), confirming that they account for most of the interaction effect.

Table 5. Per-map paired comparison of Router versus Sweeper on survival rate using Cohen's d_z (paired-sample effect size). All pairs $n = 450$, all $p < 0.001$.

Map	Mean diff. (p.p.)	Cohen's d_z	Magnitude
`bottleneck_A`	+7.71	0.81	large
`bottleneck_B`	+2.99	0.33	small
`bottleneck_C`	+2.59	0.29	small
`bottleneck_D`	+8.91	1.17	large
`open_plan_A`	+2.04	0.39	small
`open_plan_B`	+2.37	0.46	small
`open_plan_C`	+6.61	0.49	small
`open_plan_D`	+2.03	0.31	small

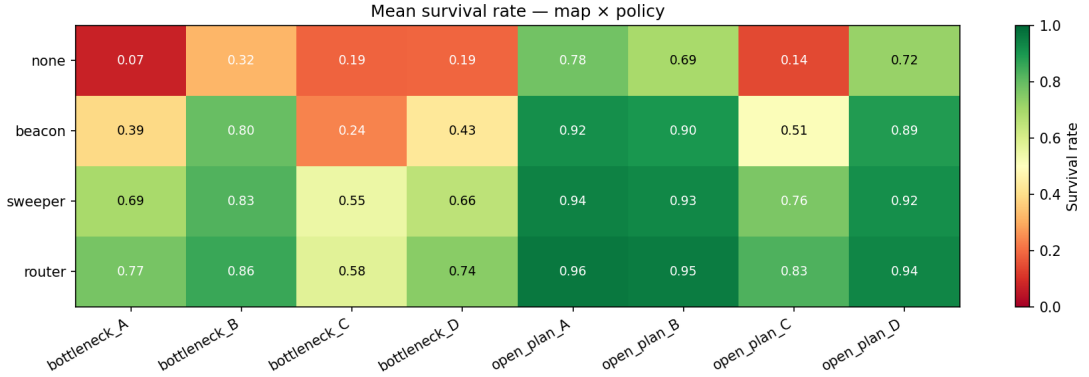


Figure 6. Mean survival rate by map and policy. Each cell is the average of 450 runs (5 fire positions × 3 crowd sizes × 30 trials).

Policy also affected the secondary Kasereka et al. metrics, though with smaller effects and different rankings from survival rate (Table 6). Mean escape time showed a medium policy effect ($\eta^2 = 0.09$), but Beacon recorded the shortest mean time (253 ticks). This reflects survivorship bias, because Beacon's smaller survivor cohort excludes the slow rescues that Router and Sweeper complete. Mean survivor potency showed a minimal effect ($\eta^2 = 0.007$), where Beacon likewise appears healthiest (73.45) for the same reason. Tukey HSD on escape time found no significant difference between Router and Sweeper ($p = 0.49$), so Router's survival advantage is not offset by slower rescues. Cumulative evacuation trajectories (Figure 7) show how evacuation unfolded over time, with Router clearing open-plan layouts earliest and holding only a small lead on the tightest bottleneck maps.

Table 6. Per-policy means for the primary outcome (survival rate) and the secondary Kasereka et al. metrics MT (mean escape time, ticks) and MP (mean survivor potency). Arrows indicate whether higher or lower is better.

Policy	Survival ↑	MT (ticks) ↓	MP ↑
none	0.388	493	71.68
beacon	0.633	**253**	**73.45**
sweeper	0.784	391	72.65
router	**0.828**	382	72.63

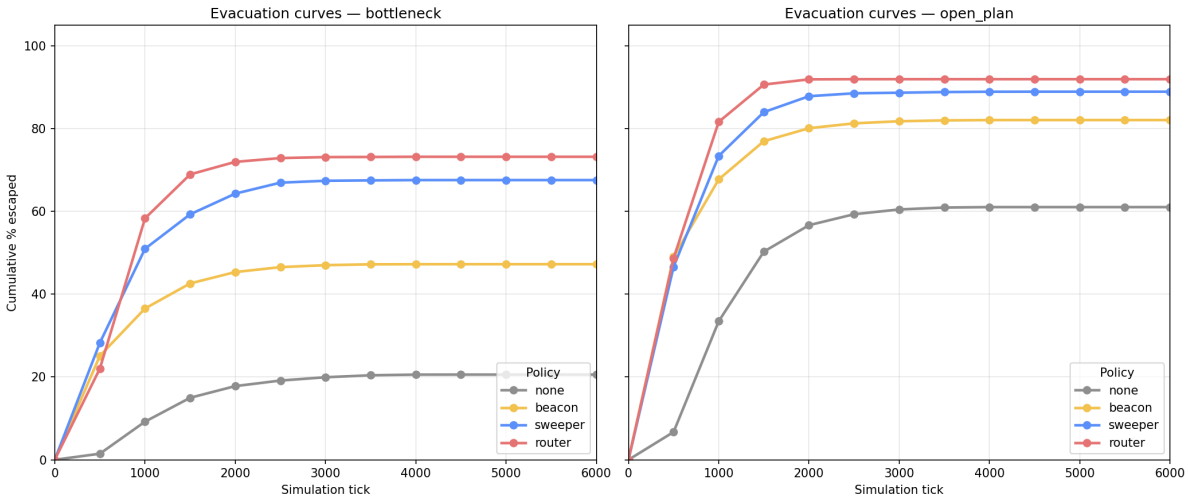


Figure 7. Cumulative evacuation over simulation time, one line per policy, split between bottleneck and open-plan map families.

6.3 Discussion

The factorial design addresses two questions: the overall ranking of the three policies, and the specific conditions under which their differences emerge. The ranking was clear, with Router placing first, Sweeper second, and Beacon third on every map tested. The gap between Router and Sweeper was small on average ($d = 0.26$), but varied fourfold across maps. The interactions are therefore central to interpreting the main ranking.

The secondary Kasereka metrics reinforce the ranking rather than qualify it. Router's survival advantage reflects genuine efficiency rather than a trade for speed. Beacon records the lowest mean escape time across all maps, but this reflects survivorship bias rather than a real trade-off, and the differences in survivor potency are negligible across policies.

Topology was the main factor shaping Router's advantage. The per-map pattern (Table 5) reveals that Router performs best where two conditions converge: congestion through shared narrow passages and multiple exit options on different walls. `bottleneck_A` combines single-cell doorways with exits on opposing walls ($d_z = 0.81$), and `bottleneck_D` directs every civilian through a central corridor before releasing them to exits at either end ($d_z = 1.17$).

Maps that remove either condition fall back to small effects. `bottleneck_B` has wide corridors that let civilian flow dissipate, leaving little congestion to reroute around. `bottleneck_C` clusters all four exits on one wall, removing the alternative routes Router's scoring would otherwise select. The four open-plan maps generally lack sustained congestion. This pattern is consistent with Shikhalev et al.'s [11] finding that congestion-aware routing approaches shortest-path when no bottleneck is present, and refines it by identifying two independent conditions that must both be present for Router to perform best.

Beacon's performance on `bottleneck_C` revealed a separate weakness tied to map geometry. All four broadcasters sit at exits, and with all exits clustered on the right wall, civilians on the left fall outside the 30 m audible range of every broadcaster. Mean survival fell to 23.8%, only 5.3 percentage points above the unguided baseline. Where the escort-capable policies retained a clear benefit on this map, passive broadcast was defeated by the exit geometry.

The crowd-size interaction was weaker but suggests a different mechanism. Router and Sweeper maintained near-constant survival across the three tested crowd sizes, while Beacon and the unguided baseline each improved by roughly 10 percentage points from the smallest to the largest. The rising trend matches the herding mechanism described by Helbing et al. [20], in which denser crowds generate more visible fleeing motion that wakes idle civilians through the crowd-direction rule. Sweeper and Router do not rely on this crowd effect, whereas Beacon's broadcast-based guidance benefits when other civilians provide visible cues.

Together, these findings answer the central research question. Router consistently achieves the highest survival across topologies. Its advantage over Sweeper depends on whether a map combines congestion with multiple exit options, and crowd size moderates the performance of passive but not active policies.

6.4 Limitations

Several limitations temper these findings. Fire and smoke rates are design parameters rather than empirically validated constants, with prior work supporting only the relative ordering of their magnitudes [16, 17]. Civilian behaviour uses a simplified herding model without Helbing's Social Force framework [3], assumes full compliance rather than the partial compliance seen in real evacuations, and assigns identical movement speeds to every civilian. Only single-floor buildings are modelled, and the eight maps serve as experimental conditions rather than a generalisation sample. Comparison against alternative leader-positioning strategies, such as the fixed-post versus walking-leader contrast in McCormack and Chen [7], was not undertaken. No real-world drill validation was performed, unlike Beyki et al. [2]. As Box [28] noted, "all models are wrong, but some are useful".

7 Conclusion and Future Work

This project compared three guide policies across a fully factorial evaluation of 14,400 simulation runs, with an unguided baseline as reference. Four findings emerge. First, guided evacuation dramatically outperformed unguided on every topology, with all guided-versus-unguided pairs in the large-effect range. Second, the three guide policies formed a monotonic ordering in which Router placed first, Sweeper second, and Beacon third, consistent across all eight maps. Third, the magnitude of Router's advantage over Sweeper varied fourfold across topologies, concentrating on maps that combined congestion with multiple exit options on different walls. Fourth, crowd size affected only the passive policies: Beacon and the unguided baseline each gained roughly ten percentage points across the tested range, while Router and Sweeper remained near-flat. The answer to the central question is therefore conditional: intelligent routing dominates all conditions tested, but how much it dominates depends sharply on the structural features of the building.

Several extensions would strengthen the work. Validation against real evacuation drills, following Beyki et al. [2], and testing on standardised benchmark layouts would together establish external validity beyond the eight hand-designed maps used here. Multi-floor environments and partial-compliance civilian models would address scope and fidelity limitations. Varying the number of guides as an additional experimental factor would capture scale effects beyond the fixed 1:25 ratio. Soft-computing alternatives such as evolutionary or reinforcement-learning policies [1] offer a complementary research direction.

References

[1] A. M. Ibrahim, I. Venkat, K. G. Subramanian, A. T. Khader, and P. De Wilde, "Intelligent evacuation management systems: A review," *ACM Transactions on Intelligent Systems and Technology*, vol. 7, no. 3, pp. 1–27, 2016.

[2] S. M. Beyki, A. S. Patricio, A. G. Lopes, A. Santiago, and L. Laím, "A modular agent-based framework for wildfire evacuation: Integrating fire spread, multimodal transport, and dynamic routing," *Safety Science*, vol. 199, 2026.

[3] D. Helbing and P. Molnár, "Social force model for pedestrian dynamics," *Physical Review E*, vol. 51, pp. 4282–4286, 1995.

[4] G. P. Senanayake, M. Kieu, Y. Zou, and K. Dirks, "Agent-based simulation for pedestrian evacuation: A systematic literature review," *International Journal of Disaster Risk Reduction*, vol. 111, p. 104705, 2024.

[5] M. Islas-Toski, E. Cuevas, M. Pérez-Cisneros, and H. Escobar, "Agent-based evacuation modeling: Enhancing building safety in emergency scenarios," *Smart Cities*, vol. 7, no. 6, pp. 3165–3187, 2024.

[6] D.-I. Curiac, O. Banias, C. Volosencu, and D. Pescaru, "Cellular automata based simulation for smoke and fire spreading in large buildings," in *International Conference on Development, Energy, Environment, Economics - Proceedings*, 2010.

[7] P. McCormack and T.-Y. Chen, "Optimizing leader proportion and behavior for evacuating buildings," in *Proceedings of the 2014 Symposium on Agent Directed Simulation*, ser. ADS '14, 2014.

[8] N. Pelechano and N. I. Badler, "Modeling crowd and trained leader behavior during building evacuation," *IEEE Computer Graphics and Applications*, vol. 26, no. 6, pp. 80–86, 2006.

[9] S. P. Hoogendoorn, F. van Wageningen-Kessels, W. Daamen, D. C. Duives, and M. Sarvi, "Continuum theory for pedestrian traffic flow: Local route choice modelling and its implications," *Transportation Research Procedia*, vol. 7, pp. 381–397, 2015.

[10] D. Shikhalev, R. Khabibulin, and A. U. Kemloh Wagoum, "Development of a safest routing algorithm for evacuation simulation in case of fire," in *Proceedings of the 6th International Conference on Agents and Artificial Intelligence (ICAART)*, vol. 1, 2014, pp. 685–690.

[11] D. Shikhalev, R. Khabibulin, U. Kemloh, and S. Gudin, "Evacuation simulation under different conditions using a safest path routing algorithm," in *Proceedings of the 18th International Conference on Enterprise Information Systems - Volume 2: ICEIS*, 2016, pp. 62–69.

[12] S.-K. Wong, Y.-S. Wang, P.-K. Tang, and T.-Y. Tsai, "Optimized evacuation route based on crowd simulation," *Computational Visual Media*, vol. 3, no. 3, pp. 243–261, 2017.

[13] Z. Zhang, H. Liu, Z. Jiao, Y. Zhu, and S.-C. Zhu, "Congestion-aware evacuation routing using augmented reality devices," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2020.

[14] D. Harel, "Statecharts: A visual formalism for complex systems," *Science of Computer Programming*, vol. 8, no. 3, pp. 231–274, 1987.

[15] J. Makmul, "A cellular automaton model for pedestrians' movements influenced by gaseous hazardous material spreading," *Modelling and Simulation in Engineering*, vol. 2020, no. 1, p. 3402198, 2020.

[16] J. Quintiere, "The spread of fire from a compartment—a review," in *Design of Buildings for Fire Safety*. ASTM International, 1979.

[17] W. Jones, R. D. Peacock, G. Forney, and P. A. Reneke, "Verification and validation of cfast, a model of fire growth and smoke spread," 2017.

[18] R. A. Brooks, "Intelligence without representation," *Artificial Intelligence*, vol. 47, no. 1–3, pp. 139–159, 1991.

[19] G. Proulx, "Occupant behaviour and evacuation," in *Proceedings of the 9th International Fire Protection Symposium*, 2001, pp. 219–232.

[20] D. Helbing, I. Farkas, and T. Vicsek, "Simulating dynamical features of escape panic," *Nature*, vol. 407, pp. 487–490, 2000.

[21] S. Kasereka, N. Kasoro, K. Kyamakya, E.-F. Goufo, P. Chokki, and M. Yengo, "Agent-based modelling and simulation for evacuation of people from a building in case of fire," *Procedia Computer Science*, vol. 130, pp. 10–17, 2018.

[22] T. Jin, "Visibility through fire smoke," *The Journal of fire & flammability*, vol. 9, pp. 135–155, 1978.

[23] F. Clark, M. Rea, and M. Ouellette, "Visibility of exit signs through smoke," 1985.

[24] C. W. Reynolds, "Flocks, herds and schools: A distributed behavioral model," *ACM SIGGRAPH Computer Graphics*, vol. 21, no. 4, p. 25–34, 1987.

[25] L. A. Zadeh, "Fuzzy sets," *Information and Control*, vol. 8, no. 3, pp. 338–353, 1965.

[26] T. Takagi and M. Sugeno, "Fuzzy identification of systems and its applications to modeling and control," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. SMC-15, no. 1, pp. 116–132, 1985.

[27] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining*, ser. KDD'96. AAAI Press, 1996, p. 226–231.

[28] G. E. P. Box, "Science and statistics," *Journal of the American Statistical Association*, vol. 71, no. 356, pp. 791–799, 1976.